SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06 **COURSE NAME:** MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I M.Ajay(192312542) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M. Ajay

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

M.Abay
R9123542

Machine learning system:

Draw, and Explain, the functional blocks of a machine learning system. Differentiate supervised and unsupervised learning with Example.

1. Functional Blocks of a Machine learning system

Scenario :- A college wants to predict, whether students will Pass or Fail.

Data collection
↓
Data preprocessing
↓
Feature Extraction
↓
Model Training
↓
Model Evaluation
↓
Final Prediction

Explanation:

1. Data collection: collect student marks, and attendance
2. Preprocessing: remove, missing, and incorrect data
3. Feature Extraction: select important features, like marks, and attendance
4. Model Training: Train the model, using collected data.
5. Evaluation: check, how, accurately the model predicts, results

supervised learning : uses labelled data such as pass and fail

unsupervised learning : uses unlabelled data to group students based on their performance.

conclusion : Machine learning learns from data and make predictions.

② Disaster Management using ML:

Design a machine learning system to predict floods and earthquakes. Explain model selection, preprocessing feature selection, training, evaluation and its uses in disaster management.

scenario : The government wants to predict floods and identify earthquake risks.

Data collection
↓
Preprocessing
↓
Feature selection
↓
Model selection
↓
Training and Evaluation

Explanation:

1. Data collection : collect rainfall, river level, earthquake records, and sensor data.
2. Preprocessing : Remove errors, fill missing values, and convert data into a suitable format
3. Feature selection : Select important features such as rainfall, waterlevel, and seismic activity.

...el Selection: use, Random, Forest, for flood-risk classification, SVM, for predicting, future, river, level. Neural Networks, can, learn, complex, patterns, from, large datasets.

5. Training and Evaluation: Train, the model, using historical data, and test, it using, new, data. Measure, its, accuracy and error, rate.

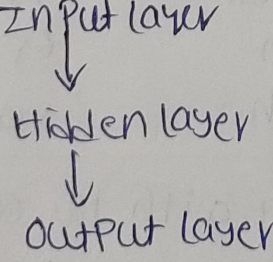
Uses:

* Early, warning
* Risk, Assessment
* Emergency Response

Conclusion: Machine, learning, helps, authorities, prepare for, disasters. Exact, earthquake, prediction, is not currently reliable.

③ Back Propagation Algorithm

Scenario: A bank, uses, a neural, network, to decide whether, a customer is eligible, for, a loan



Steps:

1- Forward Pass: Input data, passes, through, the network, to produce, a prediction.

2: Error calculation: compare, the Predicted output, with the Actual, output.

3. Background Propagation : send the error background thro the network.

4. weight update : Adjust weights to reduce the error.

5. Repeat : Repeat the process, until, the model improves

conclusion:
Back propagation reduces errors, and improves prediction, accuracy

(4) Genetic Algorithm:
Explain, how, selection, crossover, and mutation, improve candidate solution, using the given, speed, and Accuracy data set.

Scenario : A company, wants, to find the best, system, based on speed, and Accuracy.

Given Data set

Example	Speed	Accuracy	Fitness	Suitable
1.	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

1. Initial Population : start, with the five given solutions

2 selection : select, solution, with higher fitness, values such as 80 and 75.

3. crossover: combine, two selected solution, to produce a new solution.

4. Mutation : Randomly, change speed or Accuracy, to find better solution.

conclusion: The
is examples
fitness, so. T
functions.
Neural
5. sc
pre

Conclusion:

The best solution in the original dataset is examples with Medium speed, High Accuracy, and fitness 80. The best final solution depends on the fitness functions.

5. Neural Network for Disease Prediction:

Scenario: A hospital predicts disease using age, blood pressure, glucose level, and lifestyle risk.

Confusion Matrix values:

TP = 2: Disease correctly predicted

TN = 1: No disease correctly predicted

FP = 1: Disease predicted but absent

FN = 1: Disease present but missed

Accuracy: $\frac{TP+TN}{Total} \times 100 = 60\%$

Precision: $\frac{TP}{TP+FP} \times 100 = 66.67\%$

Recall: $\frac{TP}{TP+FN} \times 100 = 66.67\%$

Conclusion:

The model has 60% accuracy. Preprocessing and selecting important features can improve its performance.